



Pathways from marital relationships to children's social-emotional development in Korea: The role of parenting and social support

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ABSTRACT

Using three waves of data from the *Panel Study of Korean Children* (family $N = 1,723$), this study investigated whether and how the longitudinal effects of marital relationships on children's social-emotional development are mediated through parenting and the moderating role of social support in the mediating processes. Results showed that fathers' coparenting and warmth and mothers' parenting stress mediated the effects of marital relationships on children's externalizing and internalizing problems. Notably, some of the mediating processes were moderated by social support from extended family, friends, and neighbors. Our findings highlight the importance of considering both the proximal and broader family environment (e.g., social support from people outside the family) in understanding children's early social-emotional development.

The marital relationship is one of the most important influences on young children's development (Cox and Paley, 1997). Studies have documented concurrent and longitudinal links between marital relationships and child social, emotional, and behavioral outcomes (Sturge-Apple, Davies, & Cummings, 2010; van Eldik et al., 2020). Yet, the processes underlying the associations between marital relationships and children's social-emotional development are relatively understudied. To better support young children, it is important to understand the mediating mechanisms through which marital relationships are *longitudinally* related to children's social-emotional development and the context in which positive processes of change occur.

Grounded on the parenting process model (Schoppe-Sullivan et al., 2007), we examined the extent to which the longitudinal effects of marital relationships on children's adaptive and maladaptive development are mediated through parenting among Korean families. Examining developmental processes in non-western contexts can increase our understanding of universal as well as context-specific aspects of human development (García Coll et al., 2000). The present study used the *Panel Study of Korean Children* (PSKC) to test the mediating role of paternal and maternal parenting and the potential moderating role of social support in the hypothesized mediating processes. The PSKC is a nationally representative longitudinal study on Korean children's growth and development (see Korea Institute of Child Care and Education, 2016).

1. Pathways from marital relationships to children's social-emotional development

Interparental interactions, ranging from supporting the other and showing affection to aggressive and neglectful acts, can influence children's social-emotional development in various ways (Cowan et al., 2009). Marital relationships may have a direct impact on children's emotional and physiological systems (Davies and Woitach, 2008). Especially, exposure to destructive interparental conflict can undermine children's sense of emotional security and increase distress, avoidance, and negative appraisals (Cummings et al., 2012). Additionally, interparental interactions can serve as cognitive models of relationships for children (Bandura, 1986).

The parenting process model posits that marital relationships have indirect impacts on children through their effects on parenting (Schoppe-Sullivan et al., 2007). Specifically, emotions between parents—either positive or negative—can easily spill over into their coparenting relationship, affecting the ways and the extent to which parents work together and support each other in their parenting roles (Broderick et al., 2019; Le et al., 2016). It can also spill over into the parent-child relationship, influencing parental warmth and responsiveness (Baker et al., 2018; Han et al., 2017). Parenting stress is another aspect of parenting that can be influenced by marital relationships (Lavee, 2013). Spousal support (e.g., emotional and physical) can reduce stressors rooted in the

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parenting role (Nomaguchi and Milkie, 2020). Hostile and conflictual interactions, on the other hand, can erode parenting resources (Choi and Becher, 2019). According to the parenting process model, these aspects of parenting may serve as a mediating mechanism through which marital relationships influence children's social-emotional development.

However, previous research is limited in some ways. First, despite its compelling logic and supporting evidence in the literature, more longitudinal studies are needed. Second, studies tend to narrowly focus on a single aspect of parenting and/or the parenting of mothers. Also, many studies neither include children nor control for child effects on parenting, even though children are active contributors to their own learning and parenting (Belsky, 1984; Bronfenbrenner and Morris, 2006; Mackler et al., 2015; Yan & Ansari, 2017). Lastly, more studies are needed that include children and families in non-Western countries.

The association between marital relationships and children is indeed influenced by culturally derived values and norms (García Coll et al., 2000). Korea provides a particularly important cultural setting to examine the parenting process model. In Korean culture, family roles, relationships, and responsibilities have been rooted in the Confucian notions of family life (Rothbaum et al., 2002). One of the important Confucian family values is the parents' self-sacrifice to gain children's success (Han et al., 2017). Parents place greater emphasis on parenting responsibilities and parent-child relationships than on their own needs and interparental relationships (Kim, 2021). This raises the possibility that Korean parents might effectively compartmentalize their marital and parenting roles and keep their marital relationships from affecting their parenting and child development.

2. Differences in effects of marital relationships on maternal and paternal parenting

Some evidence indicates that the influence of marital relationships on parenting may differ between mothers and fathers (Taraban and Shaw, 2018). In the parenting literature, marital relationships often have a stronger influence on paternal parenting than maternal parenting (Lee et al., 2023; Pedro et al., 2012; Sturge-Apple et al., 2008). The gender difference may be related to the fact that mothers are typically the primary caregivers in most households (Nomaguchi and Milkie, 2020). As primary caregivers, mothers can facilitate or impede fathers' engagement in parenting (McCoy et al., 2013). When women are satisfied with their relationship with their partners, they are more likely to support and encourage fathers in parenting (Pedro et al., 2012). Another possible reason is that the male parenting role is less scripted by social norms and, thus, more susceptible to marital relationships (Nomaguchi and Milkie, 2020). Due to less clarity in the paternal parenting role, fathers might be less likely than mothers to compartmentalize and more likely to absorb marital stressors and tensions, which could lead to greater spillover to paternal parenting (Lee et al., 2023).

However, fathers' susceptibility could depend on cultural contexts (Cabrera et al., 2018). Like many Asian countries, South Korea has a patriarchal family system (Raymo et al., 2015); that is, fathers have more power and authority in the household (Kim and Wong, 2002). Under the influence of patriarchal values, marital relationships could be more important for mothers, while fathers' susceptibility to marital relationships might be muted.

3. The role of social support in parenting

Social support from people outside the family, such as extended family members, friends, and neighbors, can be an important source of support in the parenting process (Nomaguchi and Milkie, 2020; Taraban and Shaw, 2018). There are several types of social support, including emotional (e.g., esteem, encouragement), informational (e.g., advice), and instrumental (e.g., financial assistance; Armstrong et al., 2005). Social support is seen to facilitate positive outcomes by either promoting

effective adaptation processes or buffering the impact of adverse environments. Greater social support from outside the family can stimulate healthy patterns of interactions within the family and bolster parents' capacity for effective parenting. It can also buffer or attenuate the adverse influence of interparental problems on child development. The buffering hypothesis of social support suggests the presence of other supportive adults in children's lives could provide substituting resources and buffer the adverse effect of parent-related stressors (e.g., divorce, interparental conflict) on children (Armstrong et al., 2005).

In collectivistic countries such as Korea, social support is part of the everyday lives of young children and their families (Kim and Choi, 2014). Ties with the extended family generally remain very close even after marriage. Families with young children often form social networks and draw support from family, friends, and neighbors (McHale et al., 2014). Thus, social support may be a critical context to understand how family processes unfold in collectivistic societies. Previously, social support from families and friends was found to benefit the social-emotional development of children from Mexican and Dominican American backgrounds by promoting positive parenting (Serrano-Villar et al., 2017). Social support has also been found to reduce the negative impact of parents' depressive symptoms on parenting (Taraban et al., 2019). These findings suggest that social support can be an important context for children's social-emotional development. However, more research is needed to better understand the role of social support across cultural contexts.

4. The present study

Using nationally representative data, we investigated the longitudinal associations among marital relationships, parenting, and child social-emotional development among families in South Korea. First, we examined whether parenting mediates the longitudinal association between marital relationship indicators and child social-emotional outcomes. Grounded in the parenting process model (Schoppe-Sullivan et al., 2007), we expected that paternal and maternal parenting (i.e., coparenting, parental warmth, and parenting stress) would be the mediating pathways through which marital relationships influence child social-emotional development over time. Second, we examined similarities and differences between mothers and fathers. According to previous research (Cummings et al., 2004), marital relationships might have a greater impact on fathers' parenting than on mothers' parenting, suggesting that fathers' contributions to children could depend more on marital relationships. However, the patriarchal family values might lessen fathers' greater susceptibility to marital relationships. Therefore, we expected not much difference between mothers and fathers. Third, we examined whether the effects of marital relationships on children's development through parenting differed by social support from extended family, friends, and neighbors. Drawing on the social support perspective (Armstrong et al., 2005), we hypothesized that stronger social support would mitigate the harmful effects of higher marital conflict and lower marital satisfaction on children.

5. Method

5.1. Data and Sample

Data were drawn from the *Panel Study on Korean Children* (PSKC), which was conducted by the Korea Institute of Child Care and Education (KICCE). The PSKC has followed 2,150 children born in 2008, and their families, annually since 2008, with plans to continue through 2027. We used three waves of data, from child ages of 4 (Year 4; the preschool year in 2012), 6 (Year 6; kindergarten year in 2014), and 7 years (Year 7; 1st-grade year in 2015).

For the purpose of this study, we limited our sample to 1,723 children and their mothers and fathers who remained married across three waves. Excluded were children from nonintact families ($n = 46$) and

children whose parents reported that the focal child had clinically diagnosed disabilities ($n = 9$). The list of disabilities includes various physical, mental, and developmental disabilities. Lastly, 372 families with missing data on all key variables were excluded. Our study sample and original sample had only a few differences. Our sample had a higher average income ($M = 456.32$, $SD = 456.24$) than the original sample ($M = 373.24$, $SD = 445.39$). The average number of siblings was higher for our study sample ($M = 1.94$, $SD = 0.76$) than for the original sample ($M = 1.61$, $SD = 1.01$).

5.2. Measures

5.2.1. Marital Relationships

Marital Satisfaction. Marital satisfaction at Year 4 was measured using the Korean version of the Kansas Marital Satisfaction Scale (Schumm et al., 1983). The Korean version demonstrated good reliability and convergent validity in a sample of 350 Korean couples (Chung, 2004). This scale consists of four items asking respondents' satisfaction with their spouse, marriage, and marital relation (e.g., "how satisfied are you with your marriage?" and "how satisfied are you with your husband as a spouse?"). The survey was mailed to fathers and mothers, and they independently rated their satisfaction levels on a 5-point scale from 0 (*very unsatisfied*) to 4 (*very satisfied*). Parents' responses were averaged to create a composite mean score for each parent, with higher values indicating higher satisfaction ($\alpha_{\text{mother}} = .93$; $\alpha_{\text{father}} = .92$).

Marital Conflict. Marital conflict at Year 4 was measured using the Korean version of the Relationship Dynamics Scale (RDS; Markman et al., 2001). A validation study of the RDS for Korean couples found high reliability and the same factor structure as the original RDS (Chung, 2004). Example items include "little arguments escalate into ugly fights with accusations, criticisms, name-calling, or bringing up past hurts" and "my partner criticizes or belittles my opinions, feelings, or desires." Items were rated on a scale of 0 (*not at all*) to 4 (*very much*) and averaged to create a composite score for each parent ($\alpha_{\text{mother}} = .92$; $\alpha_{\text{father}} = .91$). Higher values reflect higher levels of marital conflict perceived by the respondent.

5.2.2. Parenting Processes

Parenting processes were measured using three indicators: (a) supportive coparenting, (b) parental warmth, and (c) parenting stress. At Year 6, mothers and fathers reported their own coparenting behaviors, warmth toward their child, and parenting stress using mailed questionnaires. Questionnaires were mailed to each household with the survey instructions, and parents were asked to complete them independently.

Supportive Coparenting. Supportive coparenting was measured using the family integrity subscale of the Coparenting Scale (McHale, 1997). This 7-item subscale measures parents' behaviors that actively promote the coparental unit and family integrity. The PSKC translated these items into Korean and tested the translated scale through a pilot before implementation (Choi et al., 2014). Example items were "I provide my spouse with verbal and behavioral support for their direct engagement in parent-child activities" and "I share information about the child's daily lives and positive statements with my spouse." Responses were rated on a 7-point scale (0 = *never* to 6 = *always*) and averaged to create a composite measure of supportive coparenting at Year 6 ($\alpha_{\text{mother}} = .86$; $\alpha_{\text{father}} = .88$); higher scores represent higher levels of supportive coparenting behaviors demonstrated by each parent.

Parental Warmth. Parents' warm behaviors in parent-child interactions were measured with six self-reported items. Referring to Cho and colleagues' (1999) parenting style scale, the PSKC research team developed these items. Example items include "I express affection by hugging, kissing, and holding my child," "I listen carefully to my child," and "I encourage my child to express their feelings." Responses ranged from 0 (*never*) to 4 (*always*). These items were averaged to create a

composite score for each parent at Year 6 ($\alpha = .87$ for both mothers and fathers); higher scores indicate higher levels of warmth.

Parenting Stress. Parenting stress was measured using the parenting distress subscale of the Parenting Stress Scale (PSS; Kim and Kang, 1997), which was based on the Parenting Stress Index-Short Form (PSI-SF; Abidin, 1995). This scale was tailored to assess distress related to the parenting role. Kim and Kang, 1997 found that the PSS showed adequate internal reliability and concurrent validity with the General Health Questionnaire (GHQ; Goldberg et al., 1990). The original PSS consists of 12 items, but the PSKC removed one item that was found inappropriate through a pilot test (Sin et al., 2007). Example questions include "I feel trapped by my responsibilities as a parent," "I find that taking care of my child(ren) is much more work than pleasure," and "I often feel tired, worn out, or exhausted from raising a family." Each item was rated on a 5-point scale (0 = *never* to 4 = *always*), with higher scores indicating higher parenting stress levels. These items were averaged to create a composite score for each parent at Year 6 ($\alpha_{\text{mother}} = .88$; $\alpha_{\text{father}} = .88$).

5.2.3. Child Social-Emotional Development

Child social-emotional development was measured using three indicators, including the mothers' report of externalizing and internalizing problems and the teachers' report of children's positive peer interaction skills at Year 4 and Year 7. In the Korean context, the Year 4 scores would assess children's development in preschool, and the Year 7 scores would show their development in first grade. The Year 7 scores were used as outcome variables, and the Year 4 scores were included as covariates in our analytic model.

Externalizing and Internalizing Problems. Externalizing and internalizing problem behaviors at Year 7 were measured using the Korean version of the Child Behavior Checklist (K-CBCL) for ages 6–18 (Achenbach and Rescorla, 2001; Oh & Kim, 2010). Children's previous problem behaviors at Year 4 were measured using the K-CBCL for ages 1.5–5 (Achenbach and Rescorla, 2000; Oh & Kim, 2009). In previous standardization studies (Kim et al., 2009; Oh & Kim, 2009; Oh & Kim, 2010), these two K-CBCLs demonstrated good reliability (ranging from $\alpha = .86$ to $\alpha = .89$) and validity for children in South Korea and have been used widely for research and clinical purposes.

The Year 7 externalizing scale consisted of 35 items measuring aggressive and rule-breaking behaviors; the Year 4 scale included 24 items assessing disruptive and aggressive behaviors. Also, the Year 7 internalizing scale contained 32 items, and the Year 4 scale consisted of 36 items measuring anxiety/depression, withdrawal, and somatic problems. Responses were rated on a 3-point scale (0 = *not true*, 1 = *somewhat or sometimes true*, and 2 = *very true or often true*) and summed to create externalizing problem scores at Year 4 ($\alpha = .88$) and 7 ($\alpha = .92$) and internalizing problem scores at Year 4 ($\alpha = .87$) and 7 ($\alpha = .86$). Raw scores were converted into *T* scores ($M = 50$, $SD = 10$) for comparability between scores at different waves. Higher mean scores indicate more internalizing and externalizing problems.

Positive Peer Interactions. Positive peer interaction skills at Year 7 (Grade 1) were measured using the peer interaction subscale (8 items) of the School Adjustment Inventory (Chi and Jung, 2006). Example items included "child helps other children," "child comforts others who are hurt or sad," and "helps settle peer conflicts in constructive ways." Children's peer interaction skills at Year 4 were measured using the play interaction subscale (9 items) from the Korean version of the Penn Interactive Peer Play Scale (PIPPS; Fantuzzo et al., 1998; Choi & Shin, 2008). The Korean version has established reliability and validity (Choi & Shin, 2008). The Year 4 and Year 7 scales were tailored to measure the same construct—children's positive and prosocial peer interactions in the classroom, and the correlation between the Year 4 and Year 7 scores was .25 ($p < .001$). Responses ranging from 0 (*never*) to 4 (*always*) were averaged and converted into *T* scores ($M = 50$, $SD = 10$), with higher scores denoting higher levels of positive peer interaction behaviors in the classroom ($\alpha = .78$ at Year 4 and $\alpha = .97$ at Year 7).

5.2.4. Social Support

At Year 4, mothers completed the Social Support Scale developed by Lee and Ok (2001). The PSKC team found this scale to have adequate reliability and validity with the PSKC sample (Sin et al., 2007). The scale comprises four sets of 12 items (48 items in total). Four sets were used to measure mothers' perceived levels of social support from four different sources, including mothers' own family, family-in-law, friends, and neighbors. Each set consists of 12 items designed to assess emotional, instrumental, informational, and affiliation social support. Example items include "When I feel lonely, I can open my heart and rely on them," "When I need a hand for preparing a big event, they are willing to help me," "They are willing to give me information related to child caring," and "We spend pastime or go on a vacation together." Mothers responded to the items using a 5-point Likert scale ranging from 0 (never) to 4 (always). A composite mean score was calculated for each family, with higher scores indicating greater perceived social support from people beyond the immediate family ($\alpha = .94$).

5.2.5. Covariates

Covariate selection was guided by theoretical and empirical evidence (Cabrera et al., 2014; Hotz and Pantano, 2015; Hummel et al., 2016; Ko et al., 2022; Mackler et al., 2015; Monfardini and See, 2016; Steiner et al., 2010; Taraban and Shaw, 2018; VanderWeele, 2019; Wilson and Durbin, 2010). All covariates were drawn from the Year 4 data. Family-level covariates include parents' education levels (*high school or lower degree* [reference category], *2-year college degree*, and *at least 4-year college degree*), employment status, depressive symptoms, and the number of the focal child's siblings in the household. Depressive symptoms were measured using six items (Kessler et al., 2002). Each parent's responses (ranging from 0 to 4) were averaged to create a composite score, with higher scores suggesting more severe depressive symptoms ($\alpha_{\text{mother}} = .92$; $\alpha_{\text{father}} = .92$). Parental depression significantly influences parenting and child development, especially during early childhood (Hummel et al., 2016; Taraban and Shaw, 2018; Wilson and Durbin, 2010). We controlled for depressive symptoms to adjust for its potential influences on parenting and child development. Child-level covariates include gender (1 = *female* and 0 = *male*), birth order (the oldest was coded as 1; ranging from 1 to 5; $M = 1.67$, $SD = 0.71$), externalizing and internalizing problem, and peer interaction skills at Year 4. For parenting mediators, we controlled for child social-emotional development at Year 4 to account for the effects of children on parents (Mackler et al., 2015; Yan & Ansari, 2017).

5.3. Analytic Approach

A longitudinal path analysis was conducted to examine the extent to which marital relationships are associated with children's social-emotional development directly and indirectly through its effects on parenting. Our path model has some critical elements to be noted. First, we followed a longitudinal mediation guideline (Jose, 2016) and designed a 3-time point path model. The predictor variables (i.e., marital relationship at Year 4) should occur before the mediators (i.e., parenting at Year 6), and the mediators should occur before the outcome variables (i.e., child outcomes at Year 7). Second, we adjusted child's prior skill levels to focus on development/change (Castro-Schilo and Grimm, 2017; Jose, 2016). Lastly, to account for the interdependence between the mother and the father, our paths from marital relationships to parenting include the actor and partner effects (Cook and Kenny, 2005). In longitudinal contexts, the actor effect measures a person's outcome predicted by their own past characteristics, whereas the partner effect measures a person's outcome predicted by their partners' characteristics.

As a first step, descriptive and bivariate analyses were carried out to examine study variables. Nonnormality was present in externalizing and internalizing problem scores at Year 7, as is often the case in problem behavior measures (Supplementary Table 1). To address nonnormality,

we used bias-corrected bootstrapping (BC bootstrapping). BC bootstrapping provides the most powerful and reasonable method of obtaining standard errors and confidence intervals that are robust to nonnormality (Lai, 2018; Preacher and Hayes, 2008). We also screened the bivariate correlations to check for multicollinearity (Kline, 2011). None of the correlations were above $r = .70$, indicating no issues with multicollinearity among the study variables (Supplementary Tables 2 and 3). Then, missing data analyses were conducted (Supplementary Table 4). Missing data on parent- and family-level variables ranged from 4.8% to 17.1%. Missingness on child-level variables ranged from 4.3% to 42.7%. Little's test of missing completely at random (Little, 1988) indicated that the missing data were not missing completely at random (MCAR): $\chi^2(1,509) = 2,051.12$, $p < .001$. Differences in missing data were accounted for by variables included as covariates in the model, and thus the missing at random (MAR) was presumed. We used full information maximum likelihood (FIML) to handle missing data. FIML has been shown to provide less biased estimates than other missing data techniques, especially when data do not appear to be MCAR (Allison, 2003).

To examine the first research question, a mediation model was estimated using Mplus (Muthén and Muthén, 2017). Specifically, mothers' and fathers' reports of marital satisfaction and marital conflict were entered as predictor variables (four predictors in total). Mothers' and fathers' reports of supportive coparenting, parental warmth, and parenting stress were entered as mediating variables (six mediators in total). Child externalizing problems, internalizing problems, and positive peer interaction skills at Year 7 were entered as outcome variables. Prior problem and skill levels were included as covariates to account for the stability of the outcome variables. Mediation effects were tested using the bias-corrected bootstrap method with 1,000 iterations, and bootstrapped confidence intervals were estimated (Preacher et al., 2007); the upper or lower bounds not including zero suggest a statistically significant indirect effect. Bootstrapping is also advantageous in testing mediated effects in models with more than a single mediator variable (Preacher and Hayes, 2008).

In our main analysis model, we also examined how similarly or differently marital relationships are associated with mothers' and fathers' parenting. We compared the pattern of associations between mothers and fathers. When similar associations emerged, we compared the size of parameters to test differences between the two actor effects, two partner effects, or the actor and partner effects in each role. We followed the technique of testing equality constraints (Cook and Kenny, 2005). We followed Hu and Bentler (1999) guidelines to evaluate model fit.

Lastly, we ran five moderated mediation models to examine whether the mediation effects are conditional on social support (i.e., moderated mediation). For each significant mediational path found in our path model, we added the marital relationship predictor $\times$ social support interaction term, a_3 path to parenting outcomes, as an additional predictor in the model (see Figure 1). To test simple slopes of the mediation effects moderated by social support—families with less social support ($-1SD$) and higher social support ($+1SD$), bootstrapped confidence intervals were estimated.

Table 1

6. Results

6.1. Descriptive Statistics

Table 2 shows descriptive statistics of our key variables. Mothers reported lower marital satisfaction than fathers (paired $t = -15.82$, $p < .001$), whereas their reports of marital conflict were not significantly different from fathers' reports. Also, compared to fathers, mothers reported higher levels of supportive coparenting (paired $t = 9.71$, $p < .001$), parental warmth (paired $t = 6.88$, $p < .001$), and parenting stress (paired $t = 10.74$, $p < .001$). Marital relationship and parenting

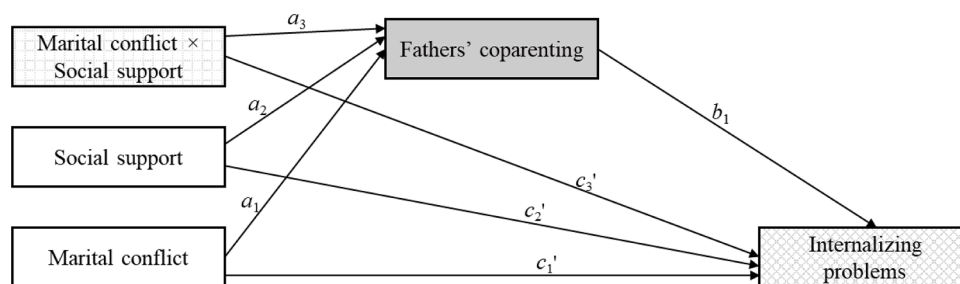


Fig. 1. An Example of a Moderated Mediation Path Diagram.

Table 1
Background Characteristics of Study Sample at Year 4.

Variables	M	(SD)	Mother	Min	Max	M	(SD)	Father	Min	Max	Paired t or McNemar
Parent characteristics											
Age (years)	34.75	(3.67)		23	51	37.20	(3.94)		23	53	-33.43***
Education, %											
High school or lower degree	29.3					26.6					5.89*
2-year college degree	26.9					19.3					35.84***
At least 4-year college degree	43.8					54.1					75.67***
Employed (18 hours or more), %	42.0					94.6					743.68***
Depressive symptoms ^a	0.94	(0.74)		0	4	0.85	(0.67)		0	4	4.43***
Family characteristics											
Household income ^b	456.32	(456.24)		0	8,150	—	—		—	—	
Number of the focal child's siblings	1.94	(0.76)		0	6	—	—		—	—	
Extended members in the household, %	9.8					—					
Child characteristics											
Age (months)	51.02	(1.23)		48	54	—	—		—	—	
Girl, %	48.6					—					
Birth order	1.66	(0.71)		1	5	—	—		—	—	
Externalizing problems ^c	49.94	(10.01)		36.67	91.54	—	—		—	—	
Internalizing problems ^d	49.94	(9.92)		36.89	102.32	—	—		—	—	
Peer interaction skills ^e	50.09	(9.88)		-3.89	72.91	—	—		—	—	

Notes. Family $N = 1,723$. Year 4 = child age of 4 years.

^aMean of 6 items rated 0 (*never*) to 4 (*always*). ^bIn Korean currency, unit of 10,000 Won (₩); ₩4,563,200 $\approx$ \$4,051 in 2012 USD. ^cMean of 24 items rated 0 (*never*) to 2 (*always*); converted to T scores. ^dMean of 36 items rated 0 (*never*) to 2 (*always*); converted to T scores. ^eMean of 9 items rated 0 (*never*) to 3 (*always*); converted to T scores.

* $p < .05$. *** $p < .001$.

Table 2
Descriptive Statistics Among the Key Variables.

Variables	Mother	(SD)	Father	(SD)	Paired t
Mother-father relationships					
(Year 4)					
Marital satisfaction ^a	2.70	(0.74)	3.02	(0.65)	-15.82***
Marital conflict ^b	1.11	(0.77)	1.08	(0.72)	1.63
Parenting (Year 6)					
Supportive coparenting ^c	4.38	(0.94)	4.11	(0.99)	9.71***
Parental warmth ^d	2.63	(0.54)	2.51	(0.59)	6.88***
Parenting stress ^e	1.57	(0.62)	1.38	(0.60)	10.74***
Child social-emotional development (Year 7)					
Externalizing problems ^f	49.99	(10.05)	—	—	
Internalizing problems ^g	50.00	(10.05)	—	—	
Adaptive school behaviors ^h	50.09	(9.93)	—	—	
Social supportⁱ (Year 4)	2.84	(0.52)	—	—	

Notes. Family $N = 1,723$. Year 4 = child age of 4 years; Year 6 = child age of 6 years; Year 7 = child age of 7 years.

^aMean of 4 items rated 0 (*very unsatisfied*) to 4 (*very satisfied*). ^bMean of 8 items rated 0 (*never*) to 4 (*always*). ^cMean of 7 items rated 0 (*never*) to 6 (*always*). ^dMean of 6 items rated 0 (*never*) to 4 (*always*). ^eMean of 11 items rated 0 (*never*) to 4 (*always*). ^fMean of 35 items rated 0 (*never*) to 2 (*always*); converted to T scores. ^gMean of 32 items rated 0 (*never*) to 2 (*always*); converted to T scores. ^hMean of 8 items rated 0 (*not true*) to 4 (*very true*); converted to T scores. ⁱMean of 48 items rated 0 (*not true*) to 4 (*very true*).

*** $p < .001$.

indicators were correlated with one another in the expected direction (Supplementary Table 2).

6.2. Effects of Marital Relationships on Child Social-Emotional Development Through Parenting

We ran a longitudinal mediation model to estimate the extent to which marital relationships are associated with children's social-emotional development directly and indirectly through parenting. This path model had a good fit, $\chi^2(36) = 68.39$, CFI = .99, RMSEA = .02, SRMR = .01. Only statistically significant paths are shown in Figure 2 (see Supplementary Table 5 for the full model findings). Our results showed no direct paths from marital relationship predictors to child outcomes were significant.

Instead, the effects of marital relationships on children were mediated through parenting (see Supplementary Table 6 for all significant mediational paths). To be specific, indirect effects via fathers' supportive coparenting behaviors were found between mothers' marital satisfaction and internalizing problems ($\beta_{\text{indirect}} = -0.01$, 95% CI [-0.02, -0.00]) and between fathers' marital satisfaction and internalizing problems ($\beta_{\text{indirect}} = -0.02$, 95% CI [-0.03, -0.00]). The results showed that marital satisfaction reported by mothers ($\beta = 0.09$, $p = .007$) and fathers ($\beta = 0.21$, $p < .001$) predicted higher supportive coparenting of fathers at Year 6, which in turn, predicted a decrease in internalizing problems at Year 7 ($\beta = -0.08$, $p = .010$). Also, an indirect effect through mothers' parenting stress was significant between fathers' marital

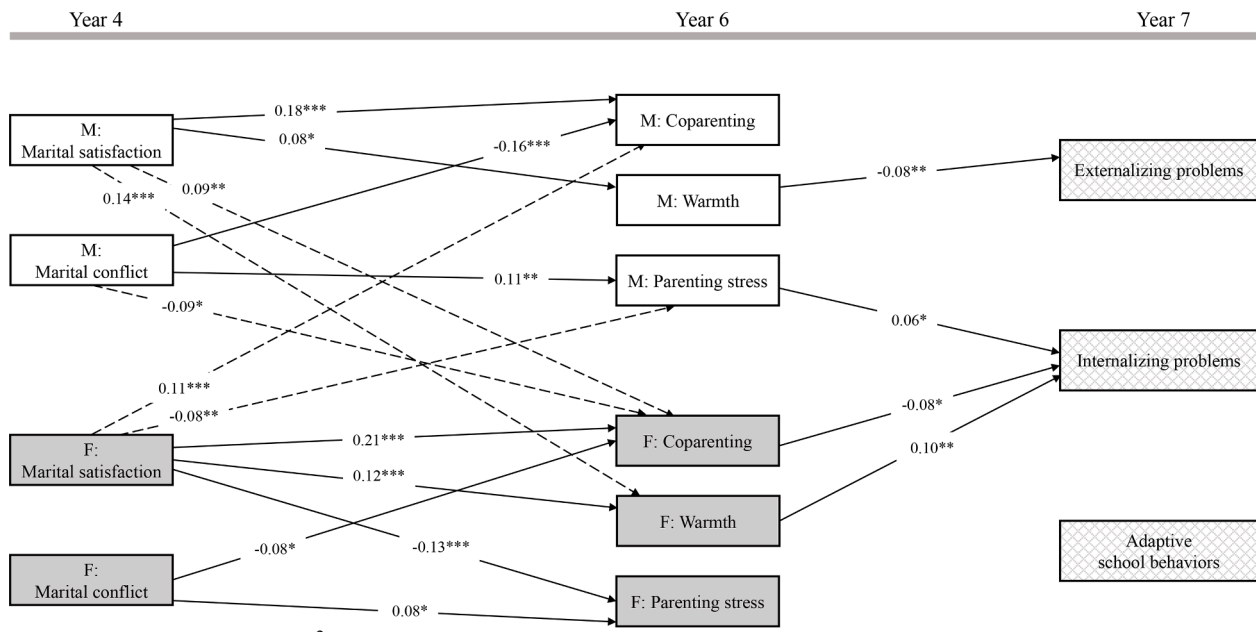


Fig. 2. 3-Wave Mediation Model Linking Mother-Father Relationships to Child's Social-Emotional Development Through Parenting

Notes. Model fit statistics: $\chi^2(36) = 68.39$, CFI = .99, RMSEA = .02, SRMR = .01. Between the mother-father relationship and parenting variables, solid lines show significant actor effects, whereas dashed lines show significant partner effects. Parents' own education, employment status, and depressive symptoms, the number of the focal child's siblings in the household, and child gender, birth order, externalizing and internalizing problems, and peer interaction skills at Year 4 were covariates for parenting mediators. Both parents' education and depressive symptoms, the number of the focal child's siblings in the household, and child gender, birth order, and prior skill levels at Year 4 were covariates for child outcomes.

* $p < .05$. ** $p < .01$. *** $p < .001$.

satisfaction and internalizing problems ($\beta_{\text{indirect}} = -0.01$, 95% CI [-0.01, -0.00]). Fathers' marital satisfaction ($\beta = -0.08$, $p = .007$) was associated with lower parenting stress of mothers at Year 6, which predicted a decrease in internalizing problems at Year 7.

Two unexpected indirect effects via fathers' parental warmth were found between mothers' marital satisfaction and internalizing problems ($\beta_{\text{indirect}} = 0.01$, 95% CI [0.01, 0.03]) and between fathers' marital satisfaction and internalizing problems ($\beta_{\text{indirect}} = 0.01$, 95% CI [0.00, 0.03]). The results showed that marital satisfaction reported by mothers ($\beta = 0.14$, $p < .001$) and fathers ($\beta = 0.12$, $p < .001$) predicted higher warmth of fathers at Year 6, which in turn, led to an increase in internalizing problems at Year 7 ($\beta = 0.10$, $p = .004$).

6.3. Effects of Marital Relationships on Maternal and Paternal Parenting

The left portion of Figure 2 shows the extent to which parents' own and their partner's perceptions of marital relationships are associated with maternal and paternal parenting (see Supplementary Table 5 for the full model findings). For both mothers and fathers, parents' own reports on marital satisfaction at Year 4 were associated with their parenting outcomes at Year 6. Specifically, higher marital satisfaction was associated with higher levels of supportive coparenting behaviors ($\beta_{\text{mother}} = 0.18$, $p < .001$; $\beta_{\text{father}} = 0.21$, $p < .001$; the size of the actor effect was similar for mothers and fathers) and parental warmth ($\beta_{\text{mother}} = 0.08$, $p = .024$; $\beta_{\text{father}} = 0.12$, $p < .001$; the size of the actor effect was similar for mothers and fathers). For fathers, higher marital satisfaction was also associated with lower parenting stress ($\beta_{\text{father}} = -0.13$, $p < .001$). In contrast, parents' own reports on marital conflict were associated with lower levels of coparenting ($\beta_{\text{mother}} = -0.16$, $p < .001$; $\beta_{\text{father}} = -0.08$, $p = .043$; no difference in the size of the actor effects between mothers and fathers) and higher parenting stress ($\beta_{\text{mother}} = 0.11$, $p = .005$; $\beta_{\text{father}} = 0.08$, $p = .035$; no difference in the size of the actor effects between mothers and fathers).

The marital relationships also showed partner effects (e.g., mothers' reports of marital relationships were associated with paternal parenting;

dotted arrows in Figure 2). Partners' marital satisfaction was associated with supportive coparenting behaviors for both mothers ($\beta = 0.11$, $p < .001$) and fathers ($\beta = 0.09$, $p = .007$); the size of these partner effects was similar for both parents. Additionally, mothers' marital satisfaction was associated with fathers' higher levels of parental warmth ($\beta = 0.14$, $p < .001$), and mothers' marital conflict was associated with fathers' lower levels of supportive coparenting behaviors ($\beta = -0.09$, $p = .029$). Lastly, fathers' marital satisfaction was associated with mothers' lower levels of parenting stress ($\beta = -0.08$, $p = .007$).

6.4. Moderated Mediation by Social Support

We found two significant moderated mediation effects (see Figure 3). First, a mediational pathway, mothers' marital satisfaction → fathers' higher coparenting → decrease in internalizing problems, was significant only for families with higher social support ($\beta_{\text{indirect for } +1SD \text{ social support}} = -0.13$, CI [-0.31, -0.03]; $\beta_{\text{indirect for } -1SD \text{ social support}} = -0.07$, 95% CI [-0.23, 0.01]; Figure 3A). Another mediational pathway, fathers' marital satisfaction → mothers' lower parenting stress → decreases in internalizing problems, was also significant only for families with higher social support ($\beta_{\text{indirect for } +1SD \text{ social support}} = -0.12$, CI [-0.30, -0.02]; $\beta_{\text{indirect for } -1SD \text{ social support}} = -0.03$, CI [-0.15, 0.01]; Figure 3B).

7. Discussion

Early child development unfolds within a family context, and the family is nested within the broader social environment (Belsky, 1984; Bronfenbrenner and Morris, 2006). In this study, we investigated the extent to which marital relationships are associated with children's social-emotional development directly and indirectly through their influence on parenting. Our analysis also took a step further to test whether social support from extended families, friends, and neighbors can moderate the effects of marital relationships on child development.

Our path model showed no direct effects of marital relationships on child social-emotional development; it is inconsistent with previous

Moderated Mediation Effects by Social Support

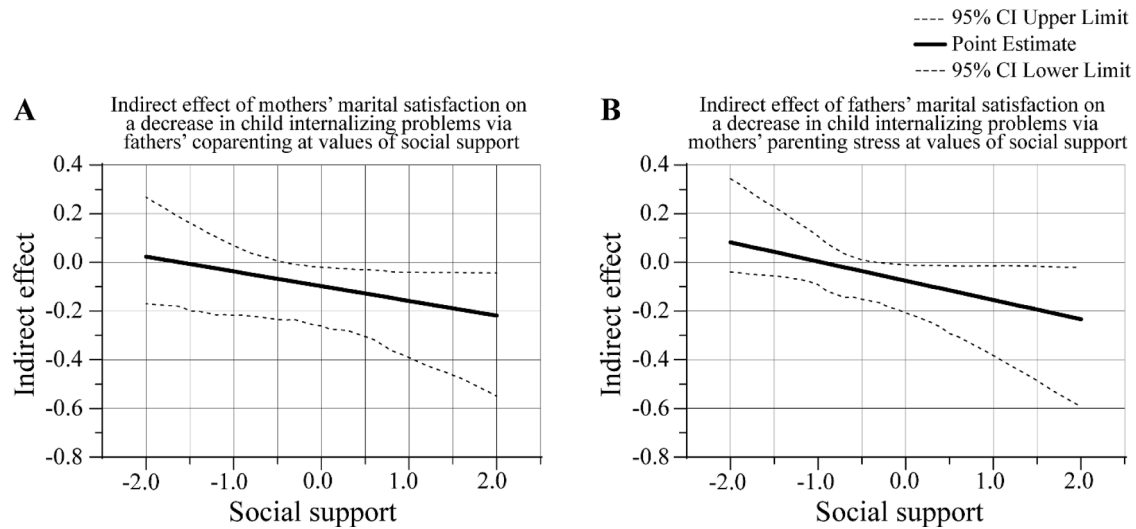


Fig. 3. Moderated Mediation Effects by Social Support

Notes. Predictor and moderator variables were centered.

Figure 3A shows the indirect effect of mothers' marital satisfaction on decreases in internalizing problems through fathers' higher coparenting was significant only for families with higher social support. Figure 3B shows the indirect effect of fathers' marital satisfaction on decreases in internalizing problems through mothers' lower parenting stress was significant only for families with higher social support.

research showing the direct link between marital relationships and child social-emotional outcomes among a sample of U.S. children (Cummings et al., 2006). Instead, several indirect pathways were found. Consistent with the parenting process model (Schoppe-Sullivan et al., 2007), the longitudinal association between marital relationships and children's social-emotional development was mediated through parenting processes. Particularly, fathers' coparenting and mothers' parenting stress emerged as the significant mediating pathways. Our results showed that parents' marital satisfaction was associated with fathers' higher levels of supportive coparenting behaviors and mothers' lower levels of parenting stress, which in turn was associated with a decrease in child internalizing problems. Fathers' coparenting and mothers' parenting stress have been found to be proximally related to both marital quality and child outcomes in Western samples (Mangelsdorf et al., 2011; Nomaguchi and Milkie, 2020; Teubert & Pinquart, 2010). Fathers' coparenting may be critically important, especially for families living in patriarchal societies (Raymo et al., 2015). Also, when the society holds gendered definitions of marriage and parenting, mothers' parenting stress might easily be affected by marital relationships and spread the influence to their children (Hong & Lee, 2019; Liu & Wang, 2015). Importantly, our moderated mediation analyses revealed that some mediational pathways were conditioned on the broader social context in which families are embedded. In a later section, we discuss mediational processes that depend on social support from people outside the family.

An unexpected finding was that parents' marital satisfaction was associated with fathers' higher warmth, which in turn was associated with increases in child internalizing problems. While seemingly counterintuitive, our results are in line with previous research. Rothenberg and colleagues (2020) found similar associations among families in cultures where parental warmth was less commonly displayed, such as in China and Thailand. One possible explanation is that fathers were increasing their warmth toward the child to reduce internalizing problems, but it may take longer for maladaptive child behaviors to respond to such higher warmth of their fathers. However, more research is needed to unpack the underlying mechanisms for this unexpected association and whether it depends on cultural contexts.

Interestingly, our results revealed the importance of positive interparental relationships for children's social-emotional development. We

found that the positive aspect of marital relationships (i.e., marital satisfaction) was associated with a decrease in child internalizing problems, whereas marital conflict was not a significant predictor of child development. Regarding the relative predictive value, negative dimensions are often expected to have larger effects than positive dimensions (van Eldik et al., 2020). However, our findings support the emerging perspective emphasizing the importance of paying more attention to positive dimensions of the marital relationship instead of an exclusive focus on marital negativity such as marital conflict and hostility (Baker et al., 2018; McHale et al., 2004).

The absence of direct and indirect associations between marital conflict and child development is worth noting. In the literature, the exclusive focus has been on the negative aspects of marital relationships. However, in our results, marital conflict was associated with their own parenting but not related to their partners' parenting or children's development. A possible explanation may come from the fact that Korean society places exceptional value on parents' investment and self-sacrifices for their children's education and success in later life (Shin et al., 2019; Woo and Hodges, 2015). This value system in Korea may have enabled parents to compartmentalize or absorb their marital distress, preventing negative emotions from spilling over and posing risks to the development of their children. Another explanation is that our analytic sample is from nationally representative data, including families with more diverse socioeconomic backgrounds, whereas many previous studies are based on low-income families and families experiencing marital difficulties. Thus, the relative importance of different dimensions of marital relationships may differ by family circumstances. Further investigation is needed to understand whether socioeconomic family circumstances change the importance of different marital dimensions.

Our moderated mediation analyses further showed that the social support from extended families, friends, and neighbors might be an important context to activate some positive mediational processes. Specifically, a mediational pathway from mother-reported marital satisfaction to a decrease in internalizing problems through higher levels of fathers' coparenting was significant only for families with higher perceptions of social support from extended families, friends, and neighbors. Another mediational pathway from father-perceived marital

satisfaction to a decrease in internalizing problems through lower levels of mothers' parenting stress was also significant only for families with higher social support. Our findings are consistent with literature showing that parents' access to social support can provide the family with important emotional and practical support (Serrano-Villar et al., 2017; Taraban et al., 2019). Social support from outside the marital dyad is likely to stimulate healthy patterns of family processes as a family asset. Our findings suggest that positive family processes may be activated through connectedness to the broader social network. One practical implication would be that interventions designed to support family functioning and child development should consider ways to increase parents' access to social support and network connections beyond the immediate family network.

Using the APIM approach, we also explored how marital relationships are similarly or differently associated with maternal and paternal parenting. Both theoretical and empirical evidence suggests that marital relationships may impact fathers' parenting more than mothers' parenting (Baker et al., 2018; Cummings et al., 2004). However, our results were mixed. Among pathways to fathers' parenting indicators, five actor effects and three partner effects emerged. Among pathways to mothers' parenting indicators, four actor effects and two partner effects were found. However, the evidence is not strong enough to conclude the gender difference between mothers and fathers. Also, there were more similarities than differences between mothers and fathers. For example, in many cases, the relative size of actor or partner effects was similar between mothers and fathers. Studies have shown that fathers and mothers are becoming more similar in many ways and marital relationships are one of the most proximal sources of support for maternal and paternal parenting (Fagan et al., 2014; Taraban and Shaw, 2018). Our findings may show similarities between parents regarding their roles and parenting in the family.

7.1. Limitations and Future Directions

This study has strengths, such as considering the broader social contexts, a focus on fathers, and a longitudinal design, but some limitations should be also noted. First, although we used longitudinal data and adjusted for a rich set of covariates, causal claims cannot be made without an experimental design. Unmeasured variables may partially account for the associations identified in this study. Carrying out replication studies in various contexts (e.g., families living with the extended family) is needed to better understand how marital relationships exert influence on children. Second, our study sample consisted of children and their biological heterosexual 2-parent families who remained intact across waves. Therefore, our findings cannot be generalized to families beyond our inclusion criteria.

Second, our sample comprised children and their parents residing in South Korea, an East Asian country. East Asian countries exhibit numerous shared characteristics, such men's and women's work and family roles rooted in the Confucian family model, traditional gender roles within marriage and child-rearing, and gendered definitions of parenting that often limit fathers' parenting (Raymo et al., 2015). Nevertheless, it is crucial to exercise caution when generalizing these findings to different East Asian family contexts. We encourage future investigations to replicate our findings within families from other East Asian countries.

Third, our study variables of marital relationships and parenting processes were based on parents' self-reports. Their responses may have been biased by the natural tendency to give socially acceptable answers. For future studies, multiple data sources would help address the social desirability biases. Also, child internalizing and externalizing problems and social support measures were assessed by mothers only. Collecting data from both parents could provide more comprehensive information. Lastly, many of our variables were measured using adapted versions of Western scales. Although the PSKC used methodological steps to translate, adapt, and cross-validate those instruments and evaluate the

psychometric properties in pilot tests, their efforts may not be enough to unpack culture-specific family processes.

Despite the limitations, our findings have significant implications for early childhood research and practice. Our results suggest that early childhood interventions/programs should pay attention to supporting interparental relationships within families, as this can have positive outcomes for parental well-being and the social-emotional development of children (Pruett et al., 2017). Additionally, our study highlights how family and broader contextual factors can jointly influence young children's social-emotional development. This implies that we need the commitment and social will to take a more comprehensive approach when designing interventions aimed at fostering positive development in young children.

CRedit authorship contribution statement

Yujin Lee: Conceptualization, Methodology, Investigation, Writing – original draft. **Kyungmin Kim:** Supervision, Writing – review & editing. **Anne Douglass:** Supervision, Writing – review & editing. **Songtian Zeng:** Supervision, Writing – review & editing.

Declaration of Competing Interest

We have no known conflict of interest to disclose.

Data availability

Data will be made available on request.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ecresq.2023.07.011](https://doi.org/10.1016/j.ecresq.2023.07.011).

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